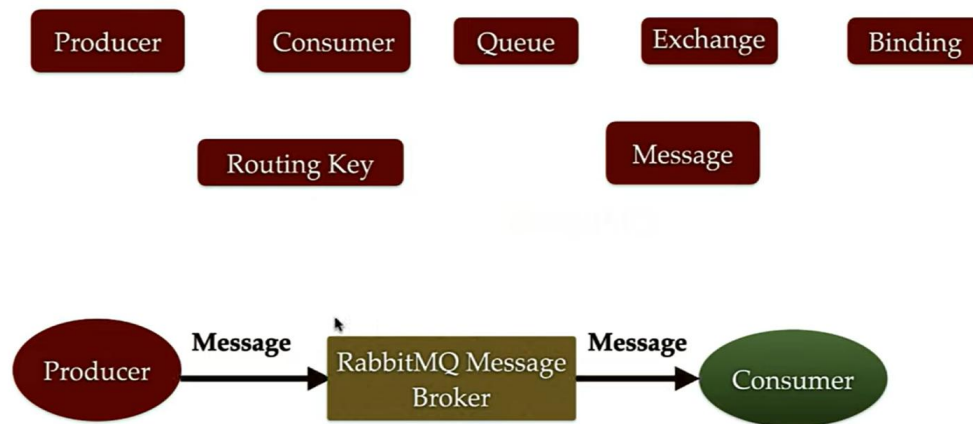
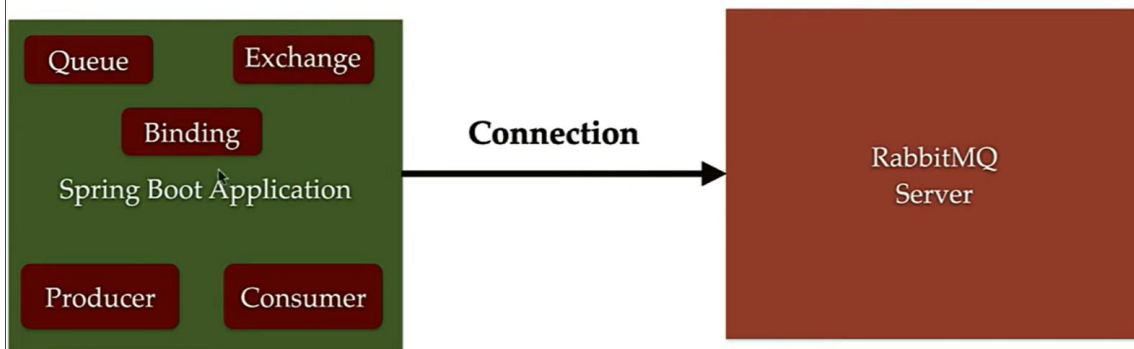


RabbitMQ Core Concepts and Architecture



Spring Boot + RabbitMQ Integration Using Spring AMQP



What is a Message Queue?

Message queuing allows applications to communicate by sending messages to each other. The message queue provides temporary message storage when the destination program is busy or not connected.

A message queue is made up of a producer, a broker (the message queue software), and a consumer.

A message queue provides an asynchronous communication between applications.



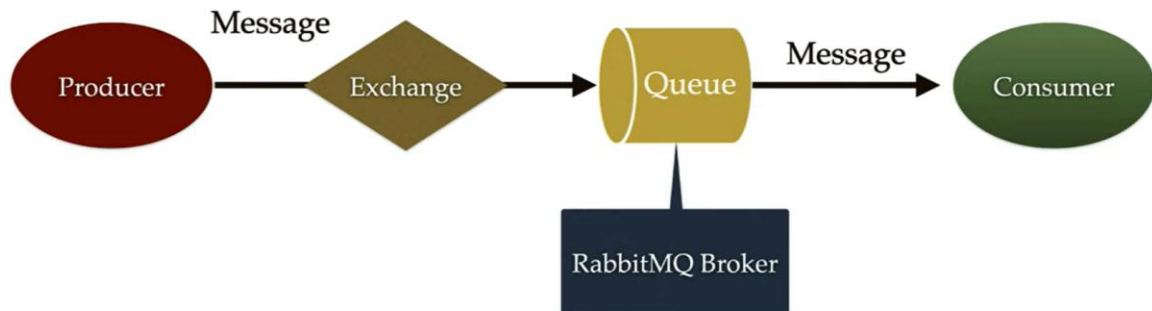
RabbitMQ

RabbitMQ is a message queue software (message broker/queue manager) that acts as an intermediary platform where different applications can send and receive messages.

RabbitMQ originally implements the Advanced Message Queuing Protocol (AMQP). But now RabbitMQ also supports several other API protocols such as STOMP, MQTT and HTTP.

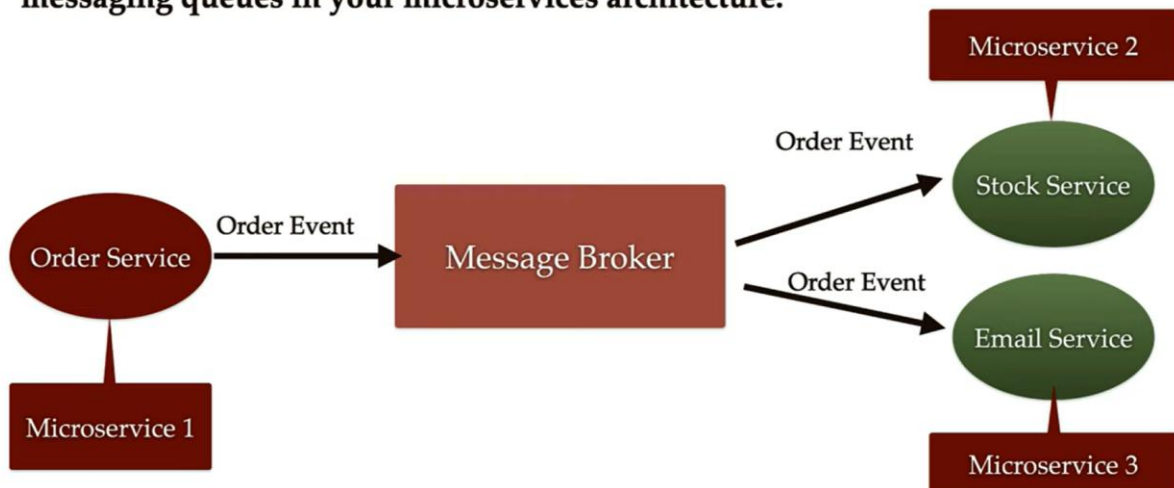
RabbitMQ

Producer is an application that sends messages to the RabbitMQ broker and
Consumer is an application that reads messages from the RabbitMQ broker.



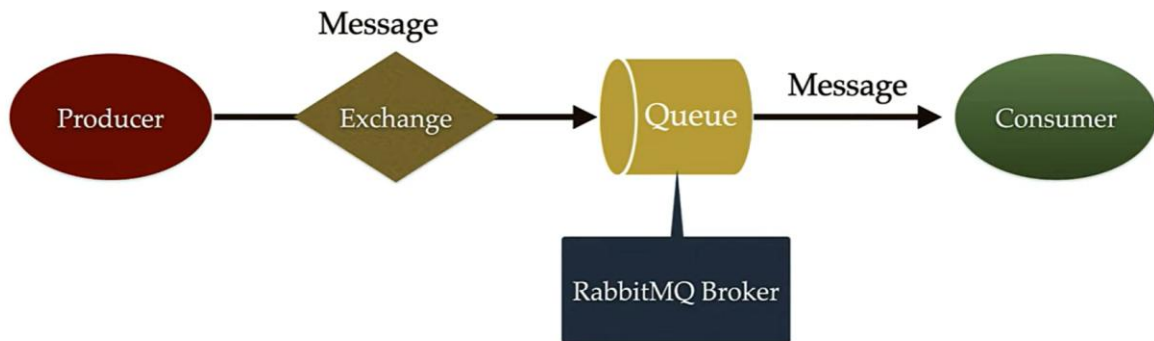
Use of RabbitMQ in Microservices

RabbitMQ is one of the simplest freely available options for implementing messaging queues in your microservices architecture.



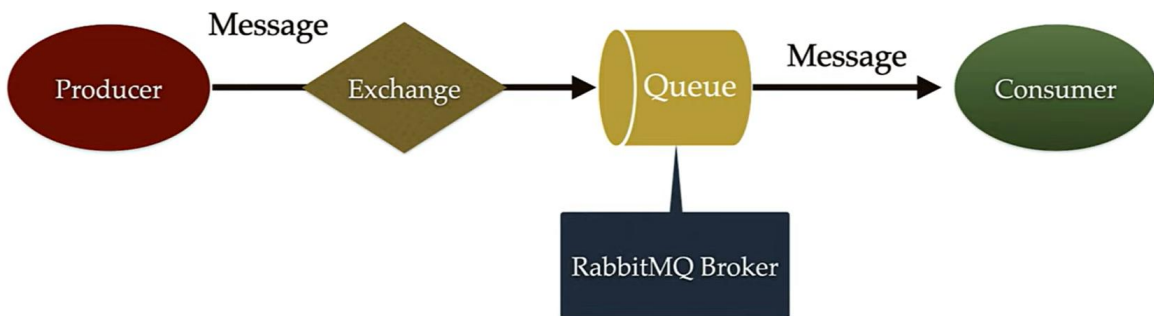
Consumer

Consumer is an application that reads messages from the RabbitMQ broker.



Producer

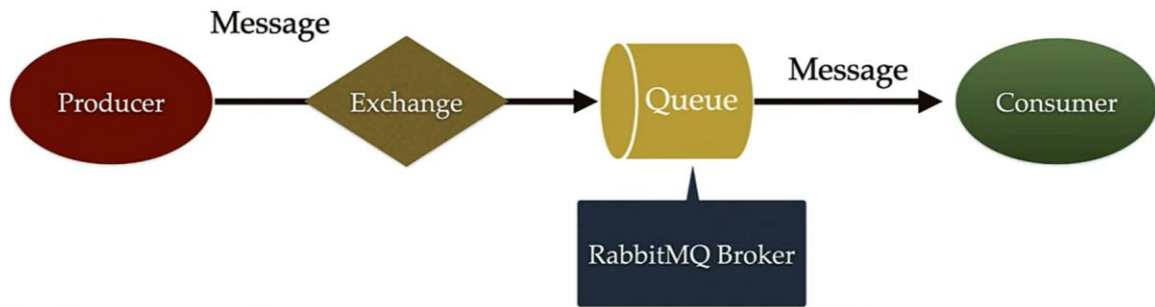
Producer is an application that sends messages. It does not send messages directly to the consumer. It send messages only to the RabbitMQ broker.



Queue

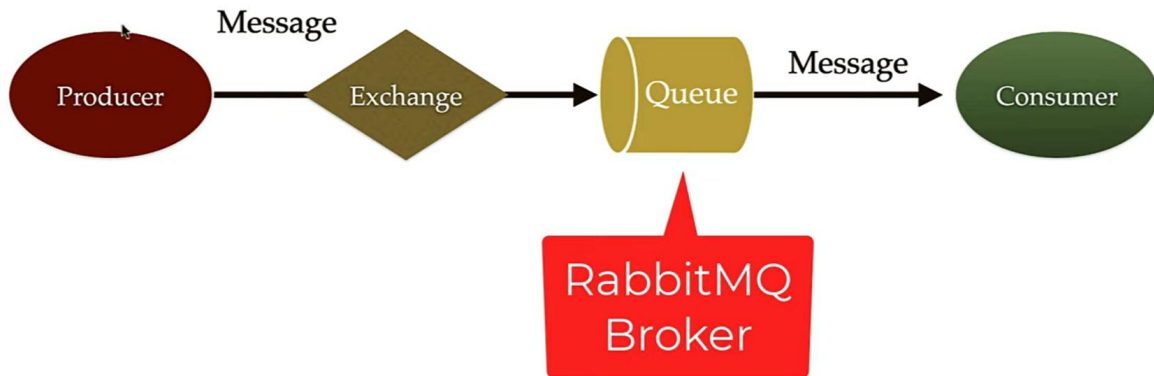
Queue is a buffer or storage in a RabbitMQ broker to store the messages.

The messages are put into a queue by a producer and read from it by a consumer. Once a message is read, it is consumed and removed from the queue. A message can thus only be processed exactly once.



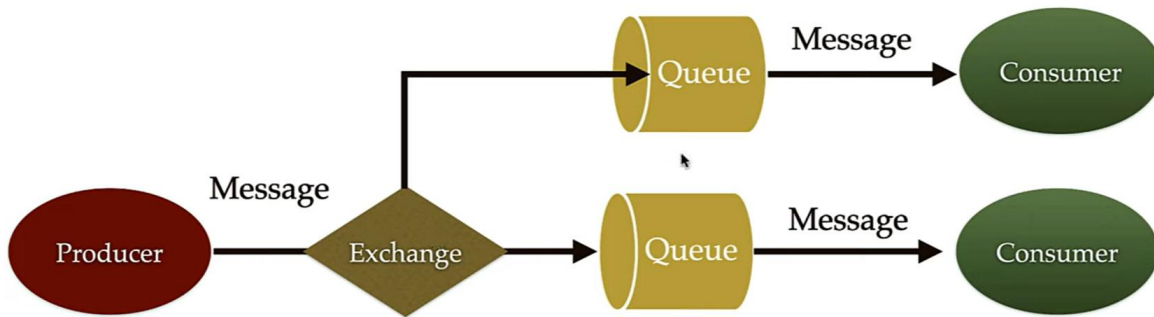
Message

Information that is sent from the producer to a consumer through RabbitMQ.



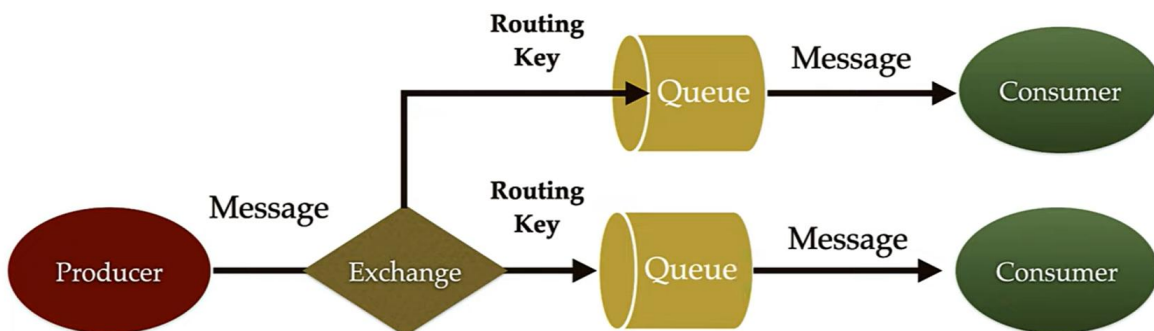
Exchange

Basically, it acts as an intermediary between the producer and a queue. Instead of sending messages directly to a queue, a producer can send them to an exchange instead. The exchange then sends those messages to one or more queues following a specified set of rules. Thus, the producer does not need to know the queues that eventually receive those messages.



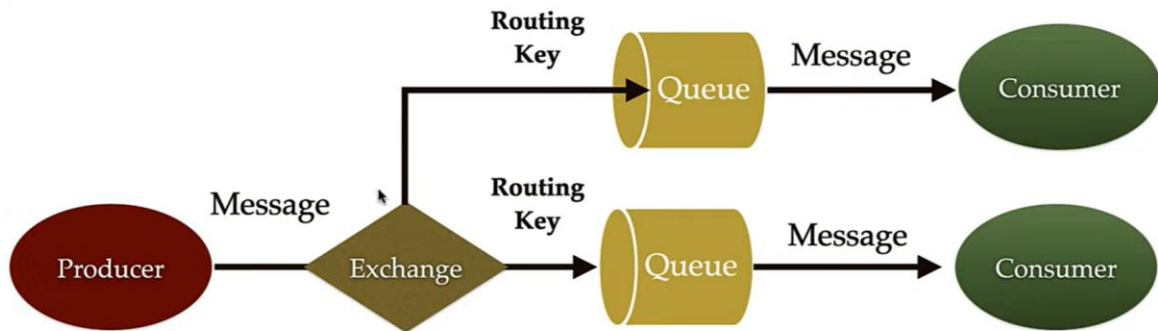
Routing Key

The routing key is a key that the exchange looks at to decide how to route the message to queues. The routing key is like an address for the message.



Binding

A binding is a link between a queue and an exchange.



RabbitMQ Architecture with Multiple Queues

